

Inclusion–exclusion and applications

Combinatorics

MATH 31003-01 - Fall 2026

The sum rule applies directly to disjoint sets. When sets overlap, ordinary addition counts some objects more than once. The principle of inclusion–exclusion repairs this overcount systematically. More importantly, it gives a general strategy for restrictions: we begin with a convenient universe, describe each way an outcome can violate the restrictions, and count intersections of those violation sets.

After studying these notes, you will be able to do the following.

- Explain the correction terms in the two-set and three-set formulas.
- State and justify the finite inclusion–exclusion formula.
- Express a desired set as the complement of a union of violation sets.
- Count bounded integer solutions by combining inclusion–exclusion with stars and bars.
- Count strings and surjective functions that use every available symbol.
- Apply block counting and inclusion–exclusion to forbidden substrings.
- Derive the formula for derangements and count permutations with a specified number of fixed points.

0.1 Correcting for overlap

If finite sets A and B overlap, the sum $|A| + |B|$ counts every element of $A \cap B$ twice: once as an element of A and once as an element of B . We subtract one copy of the intersection to count the union.

Theorem 0.1.1 (Inclusion–exclusion for two sets). For finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Example 0.1.2 (Students in two courses). In a group of students, 30 take algebra, 25 take geometry, and 8 take both. Then the number taking at least one of the two courses is

$$30 + 25 - 8 = 47.$$

Each of the first two counts includes the eight students in both courses, so subtracting eight leaves one copy of each.

For three sets, subtracting all pairwise intersections removes too much from the elements common to all three sets. Such an element is first counted three times and then subtracted three times, leaving a net contribution of zero. We must add the triple intersection once.

Theorem 0.1.3 (Inclusion–exclusion for three sets). For finite sets A , B , and C ,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

Example 0.1.4 (Students in three courses). There are 114 mathematics majors. Of these, 30 take algebra, 25 take geometry, and 50 take calculus. The three pairwise intersection sizes are 8, 15, and 12, and five students take all three courses. The number taking at least one of the courses is

$$30 + 25 + 50 - 8 - 15 - 12 + 5 = 75.$$

Consequently, $114 - 75 = 39$ take none of the three courses.

0.2 The finite inclusion–exclusion principle

For a nonnegative integer m , we write $[m] = \{1, 2, \dots, m\}$, with $[0] = \emptyset$. If $I \subseteq [m]$ is nonempty, the notation $\bigcap_{i \in I} A_i$ means that all conditions indexed by I hold simultaneously.

Theorem 0.2.1 (Principle of inclusion–exclusion). For an integer $m \geq 1$ and finite sets $A_1, A_2, \dots, A_m$,

$$\left| \bigcup_{i=1}^m A_i \right| = \sum_{\emptyset \neq I \subseteq [m]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

In words, we add the sizes of the individual sets, subtract the sizes of all pairwise intersections, add the sizes of all triple intersections, and then we continue with alternating signs.

Proof. For an element x that belongs to exactly $r \geq 1$ of the sets, the formula counts x once for every way to choose one of those r sets, subtracts it once for every way to choose two, and so on. Its total contribution is

$$C(r, 1) - C(r, 2) + C(r, 3) - \dots + (-1)^{r+1} C(r, r).$$

The binomial theorem applied to $(1 - 1)^r$ gives

$$0 = C(r, 0) - C(r, 1) + C(r, 2) - \dots + (-1)^r C(r, r).$$

Since $C(r, 0) = 1$, the first displayed alternating sum equals 1. Thus the formula counts every element in the union exactly once and counts no element outside the union. $\square$

Sometimes it is convenient to group intersections by their number of sets. We define

$$S_j = \sum_{\substack{I \subseteq [m] \\ |I|=j}} \left| \bigcap_{i \in I} A_i \right|.$$

Then inclusion–exclusion becomes

$$\left| \bigcup_{i=1}^m A_i \right| = S_1 - S_2 + S_3 - \dots + (-1)^{m+1} S_m.$$

When all j -fold intersections have the same size, S_j is that common size multiplied by $C(m, j)$.

0.2.1 The complement or violation-set form

Most applications ask for outcomes satisfying none of several bad conditions. We write U for the universe of all outcomes under consideration and A_i for the set of outcomes with violation i . The valid outcomes form

$$U \setminus \bigcup_{i=1}^m A_i.$$

Taking the complement of the union formula gives

$$\left| U \setminus \bigcup_{i=1}^m A_i \right| = \sum_{I \subseteq [m]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|,$$

where we interpret the term for $I = \emptyset$ as $|U|$.

For a restricted counting problem, we use the following workflow.

1. First, we choose a convenient universe U containing every valid outcome and whose size we can count.
2. Next, we define one violation set $A_i \subseteq U$ for each forbidden condition.
3. We state explicitly that the desired set is $U \setminus \bigcup_i A_i$.
4. For a specified index set I , we count $\bigcap_{i \in I} A_i$ by imposing all violations in I at once.
5. Finally, we sum with alternating signs and group intersections of equal size when possible.

Warning 0.2.2 (The intersections are the real work). Inclusion–exclusion is not a substitute for modeling. The formula becomes useful only after we define the universe and violation sets and learn how we can count their intersections. The formula does not require pairwise disjointness; the alternating terms correct exactly that overlap.

0.3 Application: bounded integer solutions

Translating variables removes lower bounds. Upper bounds are harder because a solution may violate several of them simultaneously. This makes bounded integer solutions a natural application of inclusion–exclusion.

Example 0.3.1 (Four bounded variables). How many integer solutions satisfy

$$x_1 + x_2 + x_3 + x_4 = 30, \quad 0 \leq x_i \leq 9.$$

Solution: We take U to be the set of all nonnegative solutions and temporarily ignore the upper bounds. Stars and bars gives

$$|U| = C(30 + 4 - 1, 4 - 1) = C(33, 3).$$

For $i \in \{1, 2, 3, 4\}$, we define

$$A_i = \{(x_1, x_2, x_3, x_4) \in U : x_i \geq 10\}.$$

The desired solutions are those in none of the four violation sets.

If j specified variables violate their upper bounds, we subtract 10 from each of them. The transformed nonnegative variables sum to $30 - 10j$, so a specified j -fold intersection has

$$C(30 - 10j + 3, 3)$$

solutions whenever $30 - 10j \geq 0$. There are $C(4, j)$ choices for the j violating coordinates. Hence the requested count is

$$\begin{aligned} & C(33, 3) - C(4, 1)C(23, 3) + C(4, 2)C(13, 3) - C(4, 3)C(3, 3) \\ &= 5456 - 7084 + 1716 - 4 = 84. \end{aligned}$$

The intersection of all four violation sets is empty because four variables cannot each be at least 10 while their sum is 30.

There is also a useful check. Replacing x_i by $9 - x_i$ gives a bijection with nonnegative solutions whose sum is $36 - 30 = 6$. The upper bounds are automatic for a sum of 6, so stars and bars gives $C(9, 3) = 84$ again.

The same argument gives a general formula.

Theorem 0.3.2 (A common upper bound). For an integer $r \geq 1$ and nonnegative integers N and q , The number of integer solutions to

$$x_1 + x_2 + \cdots + x_r = N, \quad 0 \leq x_i \leq q$$

is

$$\sum_{j=0}^{\min\{r, \lfloor N/(q+1) \rfloor\}} (-1)^j C(r, j) C(N - j(q+1) + r - 1, r - 1),$$

because larger values of j would force the transformed total to be negative.

Proof. We take U to be the set of all nonnegative solutions, with the upper bounds temporarily removed. For each $i \in [r]$, we take A_i to be the solutions in which $x_i \geq q+1$. The desired solutions form $U \setminus \bigcup_{i=1}^r A_i$. For a set $I \subseteq [r]$ of j indices, in the intersection $\bigcap_{i \in I} A_i$, we subtract $q+1$ from each variable indexed by I . The transformed variables are nonnegative and have total $N - j(q+1)$, so stars and bars gives

$$C(N - j(q+1) + r - 1, r - 1)$$

solutions when that total is nonnegative. There are $C(r, j)$ choices for I . The complement form of inclusion–exclusion gives the stated sum, and the upper limit includes exactly the feasible values of j . $\square$

If a problem has lower and upper bounds, we first translate the lower bounds. At the same time, we translate the upper bounds and then apply inclusion–exclusion to the new variables.

Problem (Practice). What inclusion–exclusion expression, before simplification, counts the nonnegative solutions of

$$x_1 + x_2 + x_3 = 20, \quad x_i \leq 8.$$

What are the universe and the three violation sets in this model?

0.4 Application: strings using every symbol

A positive requirement such as “every symbol appears” is often easier to enforce by excluding the negative events “symbol i is missing.”

Example 0.4.1 (Strings over a three-letter alphabet). How many length-6 strings over $\{a, b, c\}$ use all three letters?

Solution: We take U to be all length-6 strings over the alphabet, so $|U| = 3^6$. We take A_a , A_b , and A_c to be the strings missing the indicated letter. Each individual violation set has 2^6 elements because only two letters remain available.

If two specified letters are missing, every position must contain the one remaining letter, so each pairwise intersection has one element. No length-6 string can miss all three letters. Therefore the number using every letter is

$$3^6 - 3 \cdot 2^6 + 3 \cdot 1^6 = 540.$$

A length- n string over an m -symbol alphabet is the same object as a function from its set of positions to the alphabet. Using every symbol means that this function is surjective.

Theorem 0.4.2 (Surjections). For finite sets X and Y with $|X| = n$ and $|Y| = m$, and with the convention $0^0 = 1$, the number of surjective functions $f : X \rightarrow Y$ is

$$\sum_{j=0}^m (-1)^j C(m, j) (m - j)^n.$$

Proof. First, we assume $m \geq 1$. The universe consists of all m^n functions from X to Y . For each $y \in Y$, we take A_y to be the functions whose images omit y . If we omit j specified elements of Y , each input has $m - j$ possible images, so the intersection contains $(m - j)^n$ functions. We choose the j omitted values in $C(m, j)$ ways and use the complement form of inclusion–exclusion.

If $m = 0$, the empty function from $\emptyset$ to $\emptyset$ is surjective, while no function from a nonempty set to $\emptyset$ exists. The formula gives 1 in the first case and 0 in the second because of the convention $0^0 = 1$. $\square$

When $n < m$, a surjection is impossible. The alternating sum in the theorem is then zero. The pigeonhole principle makes that conclusion visible even though the alternating sum hides the cancellation.

0.5 Application: avoiding specified substrings

In a permutation of distinct letters, we can sometimes count the condition that a specified word appears consecutively by gluing that word into one block. Inclusion–exclusion then combines several forbidden blocks.

Example 0.5.1 (Avoiding three two-letter blocks). How many permutations of the letters in MODERNIST contain none of the substrings DO, RE, or MI?

Solution: The nine letters are distinct, so the universe U has $9!$ permutations. We define

$$\begin{aligned} A_1 &= \{\text{permutations containing DO}\}, \\ A_2 &= \{\text{permutations containing RE}\}, \\ A_3 &= \{\text{permutations containing MI}\}. \end{aligned}$$

To count A_1 , we treat DO as one object together with the other seven letters. We can arrange these eight objects in $8!$ ways. The same count holds for each individual violation.

The three prescribed blocks use disjoint letters. If two specified blocks must appear, we glue both of them and can arrange the resulting seven objects in $7!$ ways. If all three appear, the three blocks and the remaining three letters form six objects, giving $6!$ arrangements. Therefore the number containing none of the forbidden substrings is

$$9! - 3 \cdot 8! + 3 \cdot 7! - 6!.$$

Warning 0.5.2 (Blocks can interact). The gluing count above works cleanly because the three substrings use disjoint letters and have fixed internal order. If forbidden substrings overlap, share letters, or can occur more than once, an intersection may have several possible overlap patterns. Inclusion–exclusion still applies, but each intersection requires an accurate model before we insert it into the formula.

0.6 Application: derangements

Inclusion–exclusion is particularly useful when a permutation must avoid specified positions. We write a permutation of $[n]$ as a function $\pi : [n] \rightarrow [n]$. An index i is a **fixed point** if $\pi(i) = i$.

Definition 0.6.1 (Derangement). A **derangement** is a permutation with no fixed points. We denote the number of derangements of $[n]$ by D_n .

Equivalently, if n owners receive n labeled hats, phones, or exams, a derangement is an assignment in which nobody receives their own object.

Theorem 0.6.2 (Derangement formula). For every nonnegative integer n ,

$$\begin{aligned} D_n &= \sum_{j=0}^n (-1)^j C(n, j) (n-j)! \\ &= n! \sum_{j=0}^n \frac{(-1)^j}{j!}. \end{aligned}$$

Proof. We take U to be the set of all $n!$ permutations of $[n]$. For each $i \in [n]$, we define the violation set

$$A_i = \{\pi \in U : \pi(i) = i\}.$$

A derangement belongs to none of these sets.

If $I \subseteq [n]$ has size j , then every permutation in $\bigcap_{i \in I} A_i$ fixes the j specified elements. The remaining $n-j$ elements can appear in any order, so

$$\left| \bigcap_{i \in I} A_i \right| = (n-j)!.$$

There are $C(n, j)$ choices for I . Applying the complement form of inclusion–exclusion gives the first formula. Since

$$C(n, j) (n-j)! = \frac{n!}{j!},$$

factoring out $n!$ gives the second. □

The first few values are

n	0	1	2	3	4	5	6
D_n	1	0	1	2	9	44	265.

The convention $D_0 = 1$ records the one empty permutation and makes later formulas work without an exceptional case.

Example 0.6.3 (Returning six phones). Six students place their labeled phones in a box, and the phones are returned in an arbitrary order.

The number of assignments in which every student receives the wrong phone is

$$\begin{aligned} D_6 &= 6! \left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} \right) \\ &= 265. \end{aligned}$$

To count assignments in which exactly two students receive their own phones, we first choose those two students and then derange the remaining four phones. The number is

$$C(6, 2) D_4 = 15 \cdot 9 = 135.$$

The same two-stage argument proves a useful refinement.

Corollary 0.6.4 (Exactly k fixed points). The number of permutations of $[n]$ with exactly k fixed points is

$$C(n, k)D_{n-k}.$$

Proof. We choose the k fixed elements. The remaining $n - k$ elements must be deranged; otherwise there would be another fixed point. The choice of fixed elements and the derangement determine the permutation uniquely. $\square$

For instance, no permutation has exactly $n - 1$ fixed points: if all but one element are fixed, the remaining element also has nowhere else to go. The formula agrees because $C(n, n - 1)D_1 = 0$.

Dividing the derangement formula by $n!$ gives the probability that a uniformly random permutation is a derangement:

$$\frac{D_n}{n!} = \sum_{j=0}^n \frac{(-1)^j}{j!}.$$

These partial sums approach

$$e^{-1} \approx 0.367879.$$

Thus, for large n , the probability that nobody receives their own object is close to 36.8%. The expression $n!/e$ is an approximation, not the exact integer count D_n .

Problem (Practice). Eight exams are returned randomly to eight students. What are exact expressions for the number of returns in which

1. no student receives their own exam;
2. exactly three students receive their own exams;
3. at least one student receives their own exam.

0.7 Choosing useful violation sets

The four applications look different, but their solutions have the same structure.

- For bounded solutions, violation i is $x_i \geq q + 1$; shifting variables and using stars and bars counts an intersection.
- For all-symbol strings or surjections, violation i is that symbol i is missing; reducing the alphabet counts an intersection.
- For forbidden substrings, violation i is that a specified block occurs, and gluing all required blocks counts an intersection once we understand their interactions.
- For derangements, violation i is that position i is fixed, and an fixing specified positions and permuting the rest counts an intersection.

A productive violation set has two properties: the valid outcomes are precisely those with no violations, and intersections of specified violations are easier to count than the valid outcomes directly. If the intersections become harder than the original problem, a different universe or set of conditions may work better.

0.8 Common errors and a final checklist

1. **The universe.** All violation sets must be subsets of the same clearly specified set U .
2. **Precise events.** For an upper bound $x_i \leq q$, the violation is $x_i \geq q + 1$, not $x_i \geq q$.
3. **A union versus its complement.** Inclusion–exclusion starting with S_1 counts at least one violation. Starting with $|U|$ counts no violations.
4. **Simultaneous violations.** A j -fold intersection imposes all j conditions at once; it is not generally the product of the sizes of the j individual violation sets.
5. **The number of intersections.** If every j -fold intersection has a common size, we multiply it by $C(m, j)$.
6. **Impossible terms.** An empty intersection contributes zero, so we do not extend the alternating sum with meaningless binomial coefficients.

The reusable complement formula is

$$\left| U \setminus \bigcup_{i=1}^m A_i \right| = \sum_{I \subseteq [m]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|.$$

The notation is compact, but a complete solution still explains what the universe, violations, and intersections mean for the concrete objects being counted.

0.9 Further reading

The course resources page links to the books used as references for these notes. *Applied Combinatorics*, Sections 7.1–7.4, covers the inclusion–exclusion formula, surjections, and derangements. *Discrete Mathematics: An Open Introduction*, Sections 3.3 and 3.8, covers overlap bookkeeping, bounded multisets, onto functions, and derangements. *Counting Rocks!*, Section 2.7, presents the first inclusion–exclusion cases.